

LAPPEENRANTA KONNUNSUO, Finland

WMO: 027330

Lat: 61.039N

Lon: 28.564E

Elev: 46

StdP: 100.77

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.6	(c) -25.9	(d) -32.8	(e) 0.2	(f) -29.5	(g) -29.2	(h) 0.3	(i) -25.0	(j) 10.1	(k) -1.0	(l) 9.3	(m) -1.9	(n) 0.5	(o) 320	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	27.6	19.3	25.8	18.3	23.9	17.1	20.8	25.6	19.5	24.2	18.2	22.4	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	23.1	17.7	12.7	22.4	16.3	11.7	20.8	60.2	25.8	55.9	24.5	51.5	22.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.1	(b) 8.1	(c) 7.3	(e) -31.6	(f) 29.5	(g) 4.9	(h) 2.7	(i) -35.1	(j) 31.4	(k) -37.9	(l) 33.0	(m) -40.7	(n) 34.6	(o) -44.2	(p) 36.6		
(a) 9.1	(b) 8.1	(c) 7.3	(e) -32.2	(f) 22.2	(g) 4.7	(h) 1.7	(i) -35.6	(j) 23.4	(k) -38.4	(l) 24.3	(m) -41.0	(n) 25.3	(o) -44.4	(p) 26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-9.4	-7.2	-3.3	2.9	9.8	13.5	16.4	14.5	10.1	3.7	0.3	-4.2
	DBStd	9.90	8.38	7.64	5.34	3.78	4.41	3.54	3.32	3.10	3.33	3.77	4.38	6.52
	HDD10.0	2747	600	483	414	213	58	9	1	3	38	197	292	440
	HDD18.3	5268	858	716	672	462	266	149	81	124	246	453	542	698
	CDD10.0	552	0	0	0	1	52	114	198	143	42	2	0	0
	CDD18.3	31	0	0	0	0	1	4	19	6	0	0	0	0
	CDH23.3	509	0	0	0	0	41	103	263	102	1	0	0	0
	CDH26.7	93	0	0	0	0	4	15	56	18	0	0	0	0
	WSAvg	3.3	3.4	3.4	3.7	3.3	3.5	3.4	2.8	2.7	3.0	3.4	3.6	3.9
	PrecAvg	644	48	39	36	31	42	58	72	74	60	66	63	56
(15) Precipitation	PrecMax	794	79	101	65	65	143	100	163	148	139	148	131	130
	PrecMin	502	10	4	4	10	4	18	15	10	22	16	9	21
	PrecStd	78	18	21	17	14	31	20	34	38	36	31	25	24
	PrecStd	78	18	21	17	14	31	20	34	38	36	31	25	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	4.7	9.1	18.5	27.0	28.7	30.6	28.8	21.5	14.0	10.3	8.1
		MCWB	1.9	2.5	3.5	10.5	16.0	18.6	21.2	20.7	15.2	11.3	8.4	6.8
	2%	DB	2.1	2.9	6.1	15.0	24.1	26.2	28.1	26.1	19.6	12.2	8.2	5.3
		MCWB	1.3	1.7	2.6	8.4	14.8	17.7	20.4	18.6	14.5	9.9	7.0	4.0
	5%	DB	1.1	1.9	4.2	12.3	21.7	24.0	26.3	23.9	18.0	11.1	6.9	3.8
		MCWB	0.2	0.8	1.6	6.9	13.4	16.5	19.7	17.3	13.7	9.2	6.0	2.7
	10%	DB	0.4	1.0	2.8	9.8	19.4	21.9	24.2	21.8	16.5	9.9	5.6	2.4
		MCWB	-0.4	0.3	1.1	5.7	12.3	15.7	18.5	16.5	13.2	8.2	4.7	1.4
	0.4%	WB	2.2	2.6	4.4	11.0	17.5	20.9	22.9	21.7	16.9	12.1	8.5	7.0
		MCDB	2.8	3.7	7.5	17.4	24.0	25.9	27.8	27.4	20.1	13.6	9.9	8.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.5	1.8	3.1	9.3	15.8	19.0	21.4	19.5	15.6	10.5	7.0	4.2
		MCDB	2.1	2.8	5.2	13.5	21.5	24.7	26.5	24.3	18.0	11.8	8.1	5.2
	5%	WB	0.5	1.1	2.2	7.5	14.4	17.3	20.3	18.2	14.5	9.5	6.1	2.8
		MCDB	1.1	1.8	3.8	11.7	20.3	22.5	24.9	22.4	16.8	10.9	7.0	3.8
	10%	WB	-0.2	0.4	1.3	6.2	13.0	16.1	19.2	17.0	13.7	8.3	4.9	1.6
		MCDB	0.4	1.0	2.7	9.4	18.4	21.1	23.5	21.0	16.0	9.9	5.6	2.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.3	9.0	10.5	12.6	11.7	11.5	12.1	9.9	7.0	4.2	4.8
		MCDBR	3.7	5.2	9.3	15.4	17.6	15.9	15.2	16.1	12.8	9.3	5.5	4.6
		MCWBR	3.7	4.6	6.6	9.5	9.5	8.8	8.4	9.5	8.4	7.3	5.2	4.4
		MCDBR	3.5	5.0	7.6	13.3	15.0	13.9	13.8	14.3	10.4	7.6	5.2	4.4
	5% WB	MCWBR	3.6	4.6	5.7	8.7	8.8	8.2	8.1	8.8	7.3	6.2	4.9	4.3
		MCWBR	3.6	4.6	5.7	8.7	8.8	8.2	8.1	8.8	7.3	6.2	4.9	4.3
(40) Clear-Sky Solar Irradiance	taub		0.259	0.301	0.326	0.365	0.361	0.358	0.383	0.370	0.345	0.327	0.284	0.236
	taud		2.070	2.121	2.193	2.325	2.423	2.461	2.409	2.434	2.494	2.394	2.162	2.014
	Ebn at Noon		452	659	782	827	863	871	835	817	776	647	421	308
	Edh at Noon		41	74	96	103	101	100	103	93	73	58	38	26
(44) All-Sky Solar Radiation	RadAvg		0.24	0.81	2.27	3.69	5.07	5.33	5.21	4.02	2.35	0.94	0.25	0.12
	RadStd		0.04	0.19	0.31	0.40	0.59	0.35	0.49	0.46	0.25	0.13	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	-185	-178	N/A

Nomenclature: See separate page